LECTURE NOTES MATH255

SVEN SANDFELDT

1. Lecture 19

In this class we start working towards defining Galois groups. We will introduce the notion of field automorphisms and automorphisms of extensions, these will be key definitions when defining Galois groups during next class. We will focus mainly on number fields in this class, however almost everything we prove today will hold for more or less arbitrary fields (with appropriate changes to the statements).

1.1. **Automorphisms of fields and field extensions.** Let F be a field and let E be a field extension of F.

Definition 1.1. *An automorphism of a field F is an isomorphism of fields $\psi : F \to F$. We write $\mathrm{Aut}(F)$ for the set of all automorphisms of F.*

Definition 1.2. *If E is a field extension of F, then we write E/F (note that E/F is not a quotient, it is simply the notation for E being a field extension of F). We say that $\psi : E \to E$ is an automorphism of the field extension E/F if ψ is an automorphism of E such that $\psi(a) = a$ for all $a \in F$. We write $\mathrm{Aut}(E/F)$ for the set of all automorphisms of the field extension E/F. More precisely, we have:*

$$\mathrm{Aut}(E/F) = \{\psi \in \mathrm{Aut}(E) \mid \psi(a) = a, \ \text{for all } a \in F\}. \tag{1.1}$$

In the homework we will prove the following theorems:

Theorem 1.1. *Let F and E be fields, and E a field extension of F. The sets $\mathrm{Aut}(E/F)$ and $\mathrm{Aut}(F)$ are groups under composition.*

Theorem 1.2. *Let E be a field extension of F. Let $H \leq \mathrm{Aut}(E/F)$ be a subgroup, then:*

$$E^H = \{\alpha \in E \mid h(\alpha) = \alpha, \ \text{for all } h \in H\} \tag{1.2}$$

is a subfield of E that contains F, $F \leq E^H \leq E$. We call this field the fixed point field of H.

Theorem 1.3. *Let E/F be a field extension of F and let $F \leq L \leq E$ be a subfield of E that contains F. The set:*

$$H_L = \{h \in \mathrm{Aut}(E/F) \mid h(\alpha) = \alpha, \ \text{for all } \alpha \in L\} \tag{1.3}$$

defines a subgroup of $\mathrm{Aut}(E/F)$. This subgroup is called the stabilizer subgroup of L.

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Example 1.1. Let p be a prime and consider the extension $\mathbb{Q}(\sqrt{p})$ of $\mathbb{Q}$. We can define $\psi : \mathbb{Q}(\sqrt{p}) \to \mathbb{Q}(\sqrt{p})$ by $\psi(a+b\sqrt{p}) = a-b\sqrt{p}$, this map is a field isomorphism as we have shown in the homework. Moreover, if we set $b=0$ then we see that $\psi(a+0\cdot\sqrt{p}) = a-0\cdot\sqrt{p} = a$ so ψ fix every element of $\mathbb{Q}$. That is, ψ is an element of $\mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q})$.

Lemma 1.1. *Let K be a number field and let L/K be an extension of K. If $f(x) \in K[x]$ and $\alpha \in L$ is a zero for $f(x)$ then $\psi(\alpha)$ is a zero for $f(x)$ for every $\psi \in \mathrm{Aut}(L/K)$.*

Proof. Let $f(x) = a_0 + a_1x + ... + a_nx^n$ with $a_j \in K$ for $j = 0, ..., n$. If $\psi \in \mathrm{Aut}(L/K)$ and $\alpha \in L$ is a zero of $f(x)$ then we have:

$$\begin{aligned} f(\psi(\alpha)) =& a_0 + a_1\psi(\alpha) + ... + a_n\psi(\alpha)^n = \psi(a_0) + \psi(a_1)\psi(\alpha) + ... + \psi(a_n)\psi(\alpha^n) = \\ & \psi(a_0 + a_1\alpha + ... + a_n\alpha^n) = \psi(f(\alpha)) = \psi(0) = 0 \end{aligned}$$

where we have used that ψ is a homomorphism of field, so it respect multiplication and addition, and we have used that $\psi(a_j) = a_j$ for each a_j since $a_j \in K$ and ψ fix every element of the field K. ∎

Example 1.2. We can use the previous lemma to completely calculate $\mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q})$ where p is some prime. We have two elements of $\mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q})$, given by ψ from Example 1.1, and the identity map on $\mathbb{Q}(\sqrt{p})$. We claim that these are the only two elements of $\mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q})$. Indeed, let $\varphi \in \mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q})$ then $\varphi(1) = 1$ since $1 \in \mathbb{Q}$ and $\varphi(\sqrt{p}) = \pm\sqrt{p}$ since φ must map the zero $\sqrt{p}$ to some zero of the polynomial $x^2 - p$, and the only zeros of $x^2 - p$ are $\pm\sqrt{p}$. That is, we either have $\varphi(\sqrt{p}) = \sqrt{p}$ in which case φ is the identity map, or we have $\varphi(\sqrt{p}) = -\sqrt{p}$ in which case $\varphi = \psi$. So, we have $\mathrm{Aut}(\mathbb{Q}(\sqrt{p})/\mathbb{Q}) = \{\mathrm{id}_{\mathbb{Q}(\sqrt{p})}, \psi\}$.

1.2. **Conjugate algebraic numbers.** Let $\alpha \in \overline{\mathbb{Q}}$ be an algebraic number and let $f_\alpha(x) \in \mathbb{Q}[x]$ be the unique irreducible polynomial for which α is a zero. The *conjugates* of α are the other zeros of $f_\alpha(x)$.

Definition 1.3. *Let $\alpha \in \overline{\mathbb{Q}}$. The conjugates of α are the zeros of $f_\alpha(x)$. Equivalently, two elements $\alpha, \beta \in \overline{\mathbb{Q}}$ are conjugate if they are both zeros of an irreducible polynomial $f(x) \in \mathbb{Q}[x]$.*

More generally we have the notion of two elements in a field extension E/K being conjugate over K.

Definition 1.4. *Let K be a number field and E/K a finite extension of K. Two elements $\alpha, \beta \in E$ are conjugates over K if they are zeros of the same irreducible polynomial $f(x) \in K[x]$ over K.*

Lemma 1.2. *Let $\alpha, \beta \in \overline{\mathbb{Q}}$ be algebraic numbers. The numbers α and β are conjugates over a number field K if and only if the map:*

$$\psi_{\alpha,\beta} : K(\alpha) \to K(\beta), \quad \psi_{\alpha,\beta}(b_0 + ... + b_{n-1}\alpha^{n-1}) = b_0 + ... + b_{n-1}\beta^{n-1} \tag{1.4}$$

is an isomorphism. In particular, if α and β are conjugates then the map $\psi_{\alpha,\beta}$ is an isomorphism.

Proof. The fact that $\psi_{\alpha,\beta}$ is an isomorphism when α and β are conjugates is the more important direction in the lemma. We begin by showing that if $\psi_{\alpha,\beta}$ is an isomorphism, then α and β are conjugates. Let $f_\alpha(x) = a_0 + ... + a_n x^n \in K[x]$ be the irreducible polynomial for α over K. We note that if $\psi_{\alpha,\beta}$ is an isomorphism then:

$$0 = \psi_{\alpha,\beta}(f_\alpha(\alpha)) = \psi_{\alpha,\beta}(a_0 + ... + a_n\alpha^n) = a_0 + a_1\beta + ... + a_n\beta^n = f_\alpha(\beta), \tag{1.5}$$

so β is also a zero for $f_\alpha(x)$ which implies that α and β are conjugates.

We now show the more important direction, that if α and β are conjugates then $\psi_{\alpha,\beta}$ is an isomorphism. Let $f(x) \in K[x]$ be the irreducible polynomial for which both α and β are zeros. We note that the fields $K(\alpha)$ and $K(\beta)$ are both isomorphic to the field $K[x]/\langle f(x)\rangle$, and we can write these isomorphisms as:

$$K[x]/\langle f(x)\rangle \to K(\alpha), \quad p(x) + \langle f(x)\rangle \mapsto p(\alpha), \tag{1.6}$$

$$K[x]/\langle f(x)\rangle \to K(\beta), \quad p(x) + \langle f(x)\rangle \mapsto p(\beta), \tag{1.7}$$

where $p(x) \in K[x]$. It follows that we can define an isomorphism between $K(\alpha)$ and $K(\beta)$ by the composition of these isomorphisms: $p(\alpha) \mapsto p(x) + \langle f(x)\rangle \mapsto p(\beta)$, which written out is the map: $b_0 + b_1\alpha + ... + b_m\alpha^m \mapsto b_0 + b_1\beta + ... + b_m\beta^m$. That is, the map $\psi_{\alpha,\beta}$ is an isomorphism. ■

Corollary 1.3.1. *Let K be a number field. For any $\alpha \in \overline{\mathbb{Q}}$ every field homomorphsim $\psi : K(\alpha) \to \overline{\mathbb{Q}}$ that satisfies $\psi(a) = a$ for all $a \in K$ maps α to a conjugate of α over K. Conversely, for every conjugate β of α there is precisely one homomorphism $\psi_{\alpha,\beta} : K(\alpha) \to \overline{\mathbb{Q}}$, that satisfies $\psi(a) = a$ for all $a \in K$, such that $\psi_{\alpha,\beta}(\alpha) = \beta$.*

Proof. Let $f_\alpha(x) = a_0 + ... + a_n x^n$ be the irreducible polynomial for α. If $\psi : K(\alpha) \to \overline{\mathbb{Q}}$ is a homomorphism as in the lemma then we have:

$$f_\alpha(\psi(\alpha)) = a_0 + ... + a_n\psi(\alpha)^n = \psi(a_0 + ... + a_n\alpha^n) = 0 \tag{1.8}$$

so $\psi(\alpha)$ is a zero for $f_\alpha(x)$ and therefore a conjugate to α. By letting $\psi_{\alpha,\beta}$ be as in the previous lemma we see that there is some homomorphism $\psi_{\alpha,\beta} : K(\alpha) \to K(\beta) \subset \overline{\mathbb{Q}}$ that maps α to β. Conversely, if $\varphi : K(\alpha) \to \overline{\mathbb{Q}}$ maps α to β, then $\psi^{-1}_{\alpha,\beta} \circ \varphi : K(\alpha) \to K(\alpha)$ fixes α and K. However, note that any isomorphism $\eta : K(\alpha) \to K(\alpha)$ that fix both K and α satisfies:

$$\begin{aligned} \eta(b_0 + b_1\alpha + ... + b_{n-1}\alpha^{n-1}) =& \eta(b_0) + \eta(b_1)\eta(\alpha) + ... + \eta(b_{n-1})\eta(\alpha)^{n-1} = \\ & b_0 + b_1\alpha + ... + b_{n-1}\alpha^{n-1}, \end{aligned}$$

that is, any isomorphism $\eta : K(\alpha) \to K(\alpha)$ that fix K and α is the identity map. It follows that if $\varphi : K(\alpha) \to \overline{\mathbb{Q}}$ maps α to β then $\varphi = \psi_{\alpha,\beta}$ so $\psi_{\alpha,\beta}$ is the unique homomorphism mapping α to β. ■

1.3. **The extension theorem.** We end these notes by proving an important theorem which allows us to extend isomorphisms of fields to field extensions. More precisely, we have the following:

Theorem 1.4. *Let K be a number field and let $K \leq E \leq \overline{\mathbb{Q}}$ be an algebraic extension of K. If $\psi : K \to K' \leq \overline{\mathbb{Q}}$ is an isomorphism then there exists some algebraic extension $K' \leq E' \leq \overline{\mathbb{Q}}$ and an isomorphism $\tilde{\psi} : E \to E'$ such that $\tilde{\psi}|_K = \psi$. That is, we can extend ψ from K to E.*

Proof. We will only consider the case when E is a finite extension of K (this is the most important case for our purposes), for the infinite extension case we refer to the book (the proof in this case relies on Zorn's lemma). We begin by considering a simple extension $E = K(\alpha)$ of K where $\alpha \in \overline{\mathbb{Q}} \setminus K$. We will use results and notation from homework 6. Recall that we can define an isomorphism of rings $\Psi : K[x] \to K'[x]$ by:

$$\Psi(a_0 + a_1x + ... + a_nx^n) = \psi(a_0) + \psi(a_1)x + ... + \psi(a_n)x^n. \tag{1.9}$$

Let $f_\alpha(x)$ be the irreducible polynomial for α over K (not over $\mathbb{Q}$!), and recall from the homework that this implies that $\Psi(f_\alpha(x)) = g_\beta(x)$ is an irreducible polynomial over K'. Since $g_\beta(x) \in K'[x] \subset \overline{\mathbb{Q}}[x]$ and since $\overline{\mathbb{Q}}$ is algebraically closed we know that there is some $\beta \in \overline{\mathbb{Q}}$ that is a zero for $g_\beta(x)$, we fix such β. Since $f_\alpha(x)$ and $g_\beta(x)$ are irreducible over K and K' respectively we have:

$$K(\alpha) \cong K[x]/\langle f_\alpha(x)\rangle, \quad K'(\beta) \cong K'[x]/\langle g_\beta(x)\rangle. \tag{1.10}$$

Moreover, from the homework we know that the kernel of the map $\gamma \circ \Psi : K[x] \to K'[x]/\langle g_\beta(x)\rangle$ has kernel equal to $\langle f_\alpha(x)\rangle$. By the fundamental homomorphism theorem it follows that we have an isomorphism $\mu : K[x]/\langle f_\alpha(x)\rangle \to K'[x]/\langle g_\beta(x)\rangle$. Moreover, we can write this map as: $\mu(p(x) + \langle f_\alpha(x)\rangle) = \Psi(p(x)) + \langle g_\beta(x)\rangle$ so in particular, if $a \in K$, then $\mu(a) = \psi(a) + \langle g_\beta(x)\rangle$. Finally, we have isomorphisms:

$$\varphi_1 : K(\alpha) \to K[x]/\langle f_\alpha(x)\rangle, \quad b_0 + ... + b_{n-1}\alpha^{n-1} \mapsto b_0 + ... + b_{n-1}\alpha^{n-1} + \langle f_\alpha(x)\rangle,$$
$$\varphi_2 : K'[x]/\langle g_\beta(x)\rangle \to K'(\beta), \quad b_0 + ... + b_{n-1}x^{n-1} + \langle g_\beta(x)\rangle \mapsto b_0 + ... + b_{n-1}\beta^{n-1},$$

so we obtain an isomorphism $\varphi_2 \circ \mu \circ \varphi_1 : K(\alpha) \to K'(\beta)$, and this map satisfies for $a \in K$:

$$\varphi_2 \circ \mu \circ \varphi_1(a) = \varphi_2 \circ \mu(a + \langle f_\alpha(x)\rangle) = \varphi_2(\psi(a) + \langle g_\beta(x)\rangle) = \psi(a), \tag{1.11}$$

so if we let $E' = K'(\beta)$ then we have produced our extension of ψ if we let $\tilde{\psi} = \varphi_2 \circ \mu \circ \varphi_1$.

It remains to show the theorem for finite extensions E of K that are not necessarily simple. Recall that if E is a finite extension of K then there are $\alpha_1, ..., \alpha_n \in \overline{\mathbb{Q}}$ such that $E = K(\alpha_1, ..., \alpha_n)$. We begin by extending ψ to some ψ_1 defined on $K(\alpha_1)$, we can then extend ψ_1 to some ψ_2 defined on $K(\alpha_1, \alpha_2) = (K(\alpha_1))(\alpha_2)$. Proceeding by induction we obtain an extension $\tilde{\psi}$ of ψ to $E = K(\alpha_1, ..., \alpha_n)$. ■

Email address: `sandfeldt@uchicago.edu`